

Problem 1

- Need to minimize the following:

$$C(\boldsymbol{\omega}) = -\frac{1}{N}L(\boldsymbol{\omega}) + \lambda \boldsymbol{\omega}^T \boldsymbol{\omega}$$

where

$$L(\boldsymbol{\omega}) = \sum_{j=1}^n y_j(w_0 + \sum_{i=1}^p w_i x_{ji}) - \ln(1 + \exp(w_0 + \sum_{i=1}^p w_i x_{ji}))$$

We minimize this loss using gradient descent, using:

$$\boldsymbol{w}^{t+1} \leftarrow \boldsymbol{w}^t - \eta \lambda \boldsymbol{w}^t + \frac{\eta}{N} \frac{\partial L(\boldsymbol{w}^t)}{\partial \boldsymbol{w}}$$

According to the notes, we basically do an Odds ratio decision rule:

dataset	best eta	train misclass	test misclass
Gisette	0.200000	0.000000	0.020000
Madelon	1.000000	0.259500	0.426667
Dexter	0.200000	0.000000	0.140000

Table 1: Best learning rate η , training and test misclassification error, Logistic Regression.

Problem 2

We need to minimize analytically:

$$C(\boldsymbol{\omega}) = \frac{1}{N} \sum_{i=1}^n (y_i - \boldsymbol{\omega}^T \boldsymbol{x}_i - \omega_0)^2 + \lambda (\boldsymbol{\omega}^T \boldsymbol{\omega} + \omega_0^2)$$

where $\lambda = 0.001$, $(\boldsymbol{x}_i, y_i)$ are the training data, and $y_i \in \{-1, +1\}$. Predict classification via $\hat{y} = \text{sign}(\boldsymbol{\omega}^T \boldsymbol{x} + \omega_0)$.

- We can report the misclassification error of the training and test datasets in the following table:

train misclass error	test misclass error
0.000000	0.091000

Table 2: misclassification Error for training and test datasets, Ridge like regression

The train misclassification error rate is the same (0.00), though the test error rate is higher (0.091 vs. 0.020).

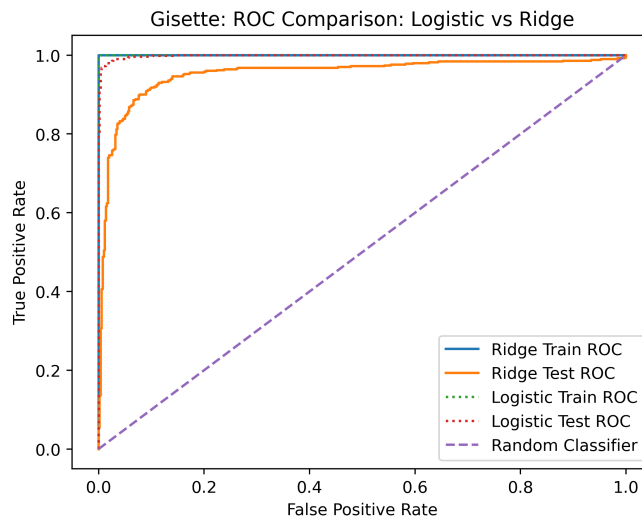


Figure 1: Average log determinant vs Lambda.

- We can display the ROC curve for both the ridge and logistic models in figure 1.

The train ROC for the logistic and ridge models are practically the same. The ROC for the logistic model is better than the ridge though; tighter to the upper left hand corner.